IILM UNIVERSITY , GREATER NOIDA , UTTAR PRADESH

A Project Report

in partial fulfilment of the requirements for the award of the Submitted degree of

BACHELOR IN TECHNOLOGY

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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Under the Supervision of

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Wildlife Classification Engine (AI/ML Project)

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Project Overview

The Wildlife Classification Engine is a CLI-based intelligent system that allows users to search animal information through a simple text-based search box. The system outputs a detailed classification including diet, lifespan, color patterns, special features, weight, and more. It is designed as a Semester 1 AI/ML project for academic learning.

The project aims to demonstrate structured data modeling, Python CLI logic, and the integration of wildlife knowledge bases.

Objectives

• Build a user-friendly CLI search engine for wildlife classification.

• Integrate structured wildlife knowledge from multiple species.

• Provide detailed animal biological characteristics.

• Lay the foundation for ML-based prediction and classification.

• Learn Python functions, loops, and dictionary-based modeling.

System Architecture

1. User enters animal name in CLI search box.

2. System checks wildlife_data dictionary.

3. If animal exists → print full classification.

4. If animal does not exist → suggest available animals.

5. Search continues until user types EXIT.

Wildlife Knowledge Base

The knowledge base includes biological and ecological details for each species:

• Category (mammal, bird, etc.)

• Diet type (carnivore, herbivore, omnivore)

• Lifespan

• Habitat type

• Color and pattern details

- Special features
- Average weight range

This structure allows easy extension of new species.

CLI Search Engine Functionality

- CLI search bar displays: "⍰ Search Animal: "
- User enters animal name.
- Engine prints complete classification.
- User can continue searching until typing EXIT.
- The system is fast, clean, and beginner-friendly.

Supported Animal Dataset

Animals currently included:

- Tiger
- Lion
- Elephant
- Panda
- Wolf

- Zebra
- Peacock

Additional animals can be added using Python dictionary expansion.

External Dataset References

Although the core classification engine uses a stored dictionary, the project references external datasets for future ML expansion:

Kaggle Wildlife Datasets:

1. African Wildlife Tracking Dataset — https://www.kaggle.com/datasets/mrisdal/wildlife-tracking

- Used for wildlife movement prediction.

2. Animals Image Dataset (for ML Classification) — Kaggle: Animals-10

- Used for CNN-based species identification (optional upgrade).

Project Code Overview

The core engine uses:

- Python Dictionary (for wildlife knowledge base)
- While-loop for continuous CLI search
- input() method for search box
- Conditional logic for classification
- Clean printing format for presentation

The code is fully compatible with:

✓Google Colab

✓VS Code

✓Windows CMD / Linux Terminal

Future Enhancements

- Image-based species detection using CNN.
- Wildlife movement prediction using GPS datasets.
- Threat detection system for forest guards.
- Animal sound recognition using audio ML.
- Full web UI using Gradio or React.
- Mobile app integration for field research.

Conclusion

The Wildlife Classification Engine successfully demonstrates a structured, intelligent, and easy-to-use wildlife information system. It is a scalable first-semester project with strong expansion potential into AI/ML applications such as image classification, sound detection, and wildlife tracking analytics.

Author Information

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Project: Wildlife Classification Engine (AI/ML)

Lets run= https://colab.research.google.com/#scrollTo=C4HZx7Gndbrh

GOOGLE COLLAB

```
# ============================================================
#  🐾 WILDLIFE CLASSIFICATION ENGINE — CLI SEARCH BOX
# ============================================================
```

```
# Wildlife Knowledge Base
wildlife_data = {
    "tiger": {
        "category": "Mammal",
        "diet": "Carnivore",
        "lifespan": "10–15 years",
        "habitat": "Tropical forests, grasslands, mangroves",
        "color_patterns": "Orange coat with black vertical stripes",
        "special_features": [
            "Strongest big cat",
            "Excellent night vision",
            "Solitary hunter"
        ],
        "average_weight": "90–300 kg"
    },
    "elephant": {
        "category": "Mammal",
        "diet": "Herbivore",
        "lifespan": "60–70 years",
        "habitat": "Forest, savannah, grasslands",
        "color_patterns": "Grey thick skin",
        "special_features": [
            "Largest land mammal",
            "Highly intelligent",
            "Uses trunk for tools and food"
        ],
```

"average_weight": "2000–6000 kg"
},
"lion": {
"category": "Mammal",
"diet": "Carnivore",
"lifespan": "12–16 years",
"habitat": "Grasslands, savannahs",
"color_patterns": "Golden coat, males have a mane",
"special_features": [
"Lives in prides",
"Powerful roar",
"Top predator"
],
"average_weight": "120–250 kg"
},
"wolf": {
"category": "Mammal",
"diet": "Carnivore",
"lifespan": "8–13 years",
"habitat": "Forests, tundra, mountains",
"color_patterns": "Grey, black or white fur",
"special_features": [
"Pack hunter",
"Strong sense of smell",
"Long-distance stamina"
],

```
    "average_weight": "25–60 kg"
  },
  "zebra": {
    "category": "Mammal",
    "diet": "Herbivore",
    "lifespan": "20–30 years",
    "habitat": "Grasslands",
    "color_patterns": "Black and white stripes",
    "special_features": [
      "Unique stripe pattern",
      "Strong herd instincts",
      "Fast runner"
    ],
    "average_weight": "200–450 kg"
  },
  "panda": {
    "category": "Mammal",
    "diet": "Omnivore (mainly bamboo)",
    "lifespan": "20 years",
    "habitat": "Mountain forests",
    "color_patterns": "Black and white fur",
    "special_features": [
      "Strong jaws for bamboo",
      "Gentle behavior",
      "Rare/endangered species"
    ],
```

```
    "average_weight": "70–120 kg"
  },
  "peacock": {
    "category": "Bird",
    "diet": "Omnivore",
    "lifespan": "15–20 years",
    "habitat": "Grasslands, farms, forests",
    "color_patterns": "Bright blue/green feathers",
    "special_features": [
      "Large colorful tail fan",
      "National bird of India"
    ],
    "average_weight": "4–6 kg"
  }
}

# ============================================================
#    SEARCH BOX (CLI LOOP)
# ============================================================
print("=========================================================")
print("      WILDLIFE CLASSIFICATION SEARCH ENGINE      ")
print("=========================================================")
print("Type an animal name and press ENTER.")
print("Type EXIT to stop.\n")

while True:
```

```
print("-------------------------------------------------------")
animal = input("🔍 Search Animal: ").lower().strip()  # CLI search box
print("-------------------------------------------------------")

if animal == "exit":
    print("\n🔚 Exiting Wildlife Search Engine. Goodbye!")
    break

if animal in wildlife_data:
    info = wildlife_data[animal]
    print(f"\n📌 RESULT: {animal.upper()}")
    print("-------------------------------------------------------")
    print(f"Category      : {info['category']}")
    print(f"Diet          : {info['diet']}")
    print(f"Lifespan      : {info['lifespan']}")
    print(f"Habitat       : {info['habitat']}")
    print(f"Color Pattern  : {info['color_patterns']}")
    print(f"Average Weight : {info['average_weight']}")
    print("\nSpecial Features:")
    for i, feat in enumerate(info["special_features"], 1):
        print(f"  {i}. {feat}")
    print("\n")
else:
    print(f"❌ '{animal}' not found in database.")
    print("Available animals:", ", ".join(wildlife_data.keys()))
    print("\n")
```

```
import numpy as np
import IPython.display as display
from matplotlib import pyplot as plt
import io
import base64

ys = 200 + np.random.randn(100)
x = [x for x in range(len(ys))]

fig = plt.figure(figsize=(4, 3), facecolor='w')
plt.plot(x, ys, '-')
plt.fill_between(x, ys, 195, where=(ys > 195), facecolor='g',
alpha=0.6)
plt.title("Sample Visualization", fontsize=10)

data = io.BytesIO()
plt.savefig(data)
image =
F"{base64.b64encode(data.getvalue()).decode()}
"
alt = "Sample Visualization"
display.display(display.Markdown(F"""![{alt}]({image})"""))
plt.close(fig)
```